DANIEL HUYNH

US Citizen | **Clearance: Top Secret** | danielhuynh523@gmail.com | (215) 870 5157 | Website: **dlhuynh.com**

EDUCATION

Johns Hopkins University, Baltimore, MD
Master of Science in Computer Science | **GPA: 4.0 / 4.0** | January 2025 – August 2026

University of Virginia, Charlottesville, VA
Bachelor of Science in Computer Science, Summa Cum Laude | **GPA: 3.92 / 4.0** | August 2021 - May 2024

EXPERIENCE

Software Engineer I
Johns Hopkins University Applied Physics Laboratory | Laurel, MD | September 2024 – Present
- Engineered **ML pipeline** for cyber alert risk prioritization on government networks as part of **2-engineer $100K+** research initiative, outperforming state-of-the-art models on multiple metrics, advancing toward **U.S. Navy adoption**.
- Lead the modernization of Dagger systems modeling software from legacy architecture to **Angular**, developing core platform features for deployment across classified government environments used by defense leadership.
- Co-own infrastructure and platform operations for a classified critical distributed system supporting **150+ total engineers**, architecting **AWS EC2** environments, **NGINX** reverse proxy infrastructure, and **Ansible** automation for scalable workflows.
- Refactor packet parsing architecture for FRISC network analysis software by replacing third-party dependencies with my own **custom packet parsing library**, reducing traffic analysis runtime by **40%**, written in **SpringBoot** with **Java**.
- Owned rebuilding of FRISC command-line tooling and session management systems while developing new network traffic analysis features for cyberattack monitoring workflows.

Software Engineer Intern
Ansys (acquired by Synopsys) | Exton, PA | May 2024 – August 2024
- Developed menu and toolbar framework for the redesigned UI of Ansys STK using **C#**, **C++**, **WPF**, and **.NET**.
- Optimized low-level backend engine functionality including application startup, file and registry I/O, and integration workflows within the STK Engine codebase using **C++**.

Software Engineer Intern
Ansys (acquired by Synopsys) | Exton, PA | May 2023 – August 2023
- Developed satellite communication conflict analysis functionality for STK, reducing computation complexity from **quadratic to logarithmic time** for **large-scale simulation workloads**, adopted by a major telecommunications company.
- Built cross-format 3D antenna model conversion tooling in **C++** to improve interoperability between external aerospace modeling formats and STK internal simulation systems.

Research Assistant, Head of IoT Team
The FloodWatch Project at UVA (**floodwatch.io**) | May 2023 - August 2024
- Led development of LiDAR-based flood mapping system using Arduino, **C++**, and **LoRaWAN** sensor infrastructure as part of an **NSF-funded** research initiative, deployed in Da Nang, Vietnam, increasing flood-map coverage by **3x**.
- Built cloud-based IoT data ingestion and processing pipelines using **AWS Lambda**, **EC2**, and **JavaScript** webhooks to collect, route, and distribute telemetry from multi-modal distributed edge sensing devices.

Perception and Motion Planning Researcher
Cavalier Autonomous Racing | April 2024 - August 2024
- Led development on LiDAR-based ground segmentation pipeline using Patchwork++ for vehicle perception in **ROS2**.
- Implemented graph **motion planning algorithms** for spatiotemporal path optimization in autonomous race environments.

***Teaching Assistant* - CS3100 (Algorithms) & CS2130 (Computer Systems) at UVA** | August 2022 – May 2023
- Led instruction and labs on algorithms, networks, runtime analysis, C/C++, Linux, and computer systems for 100+ students.

PROJECTS

BudgetBuddy* | *Capital One's Best Finance Hack Winner @ HooHacks 2023 | *JavaScript, Python, Flask, Plaid, Twilio, OpenAI*
AI financial assistant integrating Plaid transactions, budgeting analytics, & SMS chatbot workflow using Twilio and OpenAI APIs.

HealthWay | *Next.js, Typescript, FastAPI, PostgreSQL, GCP, YOLOv5*
Computer vision smart fridge platform using YOLOv5 and FastAPI to track inventory changes & generate personalized recipes.

CliniVision Project at UVA | *Next.js, Flask, PyTorch*
Computer vision medical imaging web app using CNNs & spatial transformers to classify X-ray health conditions.

SKILLS

Languages: Java, Python, C++, C#, TypeScript, JavaScript, SQL
Frameworks/Platforms: AWS (EC2, Lambda, RDS), Angular, React, Next.js, Spring Boot, PostgreSQL, .NET
Infrastructure/Systems: Linux, Git, Docker, NGINX, Ansible, ROS2, Maven, JUnit
Libraries: TensorFlow, Flask, Express, scikit-learn, PyTorch
Certifications: AWS Certified Cloud Practitioner